

Yaoru Wang

PhD Applicant · Remote Sensing and Artificial Intelligence

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RESEARCH INTERESTS

My research sits at the intersection of remote sensing and artificial intelligence. During my master's I worked on the mangroves and coastal waters of Hainan, addressing three problems: high-resolution image reconstruction, long-term change detection, and fine-grained species classification, while exploring how quantum computing can be applied to the intelligent interpretation of remote sensing imagery. For my PhD I hope to continue in coastal-zone remote sensing, ecological remote sensing, or intelligent processing of remote sensing images.

EDUCATION

Hainan Normal University · College of Artificial Intelligence · Computer Technology (Electronic Information)

Sep 2024 - Jun 2027 (expected)

Master's candidate · Advisor: Shulei Wu

Research focus: mangrove monitoring and species classification, satellite-derived bathymetry, deep learning for remote sensing

Hainan Normal University · Computer Science and Technology

Sep 2020 - Jun 2024

Bachelor of Engineering · GPA 2.67

RESEARCH EXPERIENCE

Master's thesis project: long-term reconstruction and dynamic monitoring of mangrove habitat trajectories

Sep 2025 - present

College of Artificial Intelligence, Hainan Normal University · independent work

- Built a frequency-domain refinement adversarial network (Camelot-Net) that suppresses tidal, cloud-shadow and phenological noise in multi-year time series and removes spurious change points from per-pixel trajectories.
- Introduced spatial-texture constraints along the time axis to fix the fragmentation in space and the abrupt jumps in time produced by pixel-level methods.
- Completed the network and the island-wide experimental design for Hainan; the work is being prepared as a manuscript.

Master's thesis project: fine-grained mangrove species classification with a hybrid quantum-classical neural network

Sep 2025 - present

College of Artificial Intelligence, Hainan Normal University · independent work

- Designed a hybrid quantum-classical network coupling red-edge feature enhancement with a variational quantum circuit to separate spectrally similar species such as *Kandelia obovata* and *Rhizophora stylosa*.
- Implemented quantum state encoding and variational quantum classifiers in Qiskit and QPanda, trained jointly with a classical convolutional feature extractor.
- Built the multi-species sample set for the Dongzhai Harbor National Nature Reserve; the work is being prepared as a manuscript.

Lightweight shrimp segmentation and edge computing for complex aquaculture environments Oct 2025 - Oct 2026

Industry-topic project, National Innovation and Entrepreneurship Education Practice Base (Banyan Fund) · project leader

- Built a shrimp segmentation dataset of more than 8,000 images covering day, overcast and night conditions and clear versus turbid water, with a unified annotation and quality-control protocol.
- Designed a lightweight MobileViTv3 network with multi-scale dilated convolutions and ECA channel attention; after structured pruning and INT8/FP16 quantisation it runs above 20 FPS and below 15 W on a Jetson edge device.

Remote sensing estimation of rubber plantation above-ground biomass and phenology

2025

2025 Science and Technology Project, Hainan State Farms Investment Holding Group · contributor

- Built a 170-dimensional feature set from Sentinel-2 imagery and 64 field plots, combining spectral bands, vegetation indices and GLCM texture; introduced an NDTI texture index with PCA and evaluated biomass accuracy by R2, RMSE and MAE.

Spatio-temporal accounting of mangrove blue carbon and future climate scenario simulation using multimodal AI and remote sensing

2025

National Natural Science Foundation of China proposal · co-writer

- Helped design the four-stage roadmap - long-term dynamic mapping, multimodal AI blue-carbon accounting, spatio-temporal carbon accounting and future climate scenario simulation - for Hainan mangroves over 1998-2024.

Key technologies for monitoring and assessing mangrove blue carbon with multi-source remote sensing 2025

Hainan Provincial Natural Science Foundation key project · contributor

- Contributed to the project rationale and the literature review, and proposed a change detection network based on CNN and Transformer residual feature extraction.

Integrated communication, navigation and remote sensing equipment for marine fishery applications (feasibility study)

2025

Space-air-ground-sea programme · co-writer

- Co-wrote the feasibility study report, reviewing integrated communication, navigation and remote sensing for fishery resource monitoring, vessel supervision and smart aquaculture early warning, and compiling technical indicators for maritime risk perception and vessel avoidance.

National Undergraduate Innovation and Entrepreneurship Training Program

2022

Project leader · national-level project, completed

Quantum Computing and Programming course projects

2025

independent work

- Near-Earth object classification: implemented and compared a quantum naive Bayes classifier with a classical Gaussian naive Bayes model, evaluating accuracy and positive/negative-class detection and visualising quantum state measurements.
- Quantum clustering: compared classical KMeans with fidelity-based quantum KMeans in clustering quality, convergence and computational cost.

PUBLICATIONS

- [1] Wang Y., Zhang A., Wu S., Wu Z. Application of the Transformer model to Sentinel-2 data for accurate water depth estimation in a coastal environment. Remote Sensing Letters, 2026, 17: 1169-1185.

First author · JCR Q4 · DOI: 10.1080/2150704X.2026.2687826

- [2] Zhao Y., Wang Y., Zhang A., Liu G., Wu S., Chen H. Cross-domain mangrove change detection and ecological response analysis under typhoon disturbance. *Journal of Environmental Management*, 2026, 397: 128283.

Second author · JCR Q1

- [3] Liu G., Wu S., Wang Y., et al. Mangrove taxon mapping from Sentinel-1/2 using a compact one-dimensional network with probability calibration, predictive entropy, and frozen transfer evaluation. *Frontiers in Plant Science*, 2026, in press.

Third author · JCR Q1 · accepted

- [4] Fu L., Wang Y., Wu S., et al. TCCFNet: a semantic segmentation method for mangrove remote sensing images based on two-channel cross-fusion networks. *Frontiers in Marine Science*, 2025, 12: 1535917.

Second author · JCR Q1

- [5] Chen H., Wang Y., Liu G., Wu S., Chen H., Wu Z. Spectral-geospatial XGBoost regression for bathymetric mapping in the Yellow River. *Marine Geodesy*, 2025: 1-34.

Second author · JCR Q3

- [6] Wu S., Zhao Y., Wang Y., Chen J., Zang T., Chen H. Convolution feature inference-based semantic understanding method for remote sensing images of mangrove forests. *Electronics*, 2023, 12(4): 881.

Third author · JCR Q3

- [7] Zhao Y., Wu S., Wang Y., Chen H., Zhang X., Zhao H. Fire detection algorithm based on an improved strategy of YOLOv5 and flame threshold segmentation. *Computers, Materials & Continua*, 2023, 75(3): 5639-5657.

Third author · JCR Q3

MANUSCRIPTS UNDER REVIEW

- [1] LASGM: learnable adaptive spectral-group masking for multispectral masked autoencoder pretraining in mangrove segmentation.
- [2] Wald-consistent super-resolution of Sentinel-2 red-edge and SWIR bands via endogenous NIR structure guidance and actor-ADMM optimization.
- [3] A unified spatio-temporal framework enables robust monitoring of the changes in mangroves in the vast tropical coastal areas of southern China.
- [4] Remote sensing bathymetry retrieval method based on quantum-classical dual-channel feature fusion.
- [5] A frequency-domain enhanced U-Net method for underwater depth inversion under complex optical conditions.
- [6] Optimizing masking policies via deep reinforcement learning in MAE for robust hyperspectral image classification. Manuscript under review.

PATENTS AND SOFTWARE COPYRIGHTS

- [1] Time-series change detection of mangroves based on a multi-scale residual GRU network. Invention patent, 2026.

Third inventor

- [2] Water depth inversion method based on particle swarm optimization and dynamic segmented ensemble learning. Invention patent, 2025.

Fourth inventor

- [3] Body temperature detection and trajectory tracking method based on face recognition. Invention patent, 2022.
Fifth inventor
- [4] Machine-learning-based epidemic prevention and detection system for local areas. Software copyright, 2021.
Fifth contributor

CONFERENCES AND TALKS

- [1] The 2nd National Coastal Zone Conference and Intelligent Coastal Zone Forum (2024, attendee)
- [2] The 15th National Symposium on Biodiversity Science and Conservation (2024, attendee)
- [3] The 6th International Conference on Computer Science and Educational Informatization (2024, attendee)
- [4] The 3rd National Coastal Zone Conference (2025, attendee)
- [5] The 37th Symposium on Marine Surveying and Mapping (2025, attendee)
- [6] The 1st Conference on Coastal Marine Resources and Environment (2025, oral presentation)
- [7] International Marine Science and Technology Conference and the 2025 International Marine Science and Technology Equipment Exhibition (2025, attendee)

HONORS AND AWARDS

The 15th Chinese Collegiate Computing Competition · provincial first prize	2022
"Jiuqi Nuwa Cup" Hainan Provincial Low-Code Programming Competition · provincial first prize	2023
The 7th China International "Internet+" College Students' Innovation and Entrepreneurship Competition · university first prize	2021
Hainan Normal University Undergraduate Innovation and Entrepreneurship Training Program · university first prize	2021
Chinese Collegiate Computing Competition · university first prize	2023
The 14th "Higher Education Press Cup" National College Mathematical Contest in Modeling · provincial second prize	2023
China Collegiate Computing Contest · university second prize	2023
"Higher Education Press Cup" National College Mathematical Contest in Modeling · provincial third prize	2021
Academic Scholarship, College of Artificial Intelligence, Hainan Normal University · second class	2026
Academic Scholarship, College of Artificial Intelligence, Hainan Normal University · third class	2026
Academic Scholarship, College of Information Science and Technology, Hainan Normal University	2023

SKILLS AND LANGUAGES

Programming and tools: Python (PyTorch, TensorFlow, OpenCV, scikit-learn), ENVI, ArcGIS / QGIS, Google Earth Engine, Qiskit / QPanda, Git, Linux and GPU training environments

Remote sensing: acquisition, preprocessing and atmospheric correction of Sentinel-1/2, Landsat, ICESat-2 / GEDI, hyperspectral and UAV imagery; feature engineering, model development and accuracy assessment

Languages: IELTS 6.5 (August 2026)